

Lecture 1-4

→ Introductions

- Supervised & unsupervised learning
- ↳ Regression and classification

#

→ Linear Regression

- hypothesis & θ update rule: gradient descent
- gradient descent: batch v/s stochastic
- normal equation & derivation → properties of trace derivatives

→ Locally Weighted Regression

- weight function definition
- intro to distribution rotations

→ Why we use least squares in linear regression (★)

→ Logistic Regression

- new hypothesis with same θ update rule
- Newton's method for scalar θ
- Newton's method for vector θ (★)

→ Perceptron Algo

• definition, bernoulli, gaussian

→ Exponential Families

→ General Linear Models • definition, blueprint & training

→ Softmax Regression (★)